Customer Support Assistant (NLP + RL)

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1. PROBLEM STATEMENT

A lot of companies today receive a huge number of customer messages every day. Answering all of these manually takes too much time and is expensive. Simple chatbots also exist, but they usually only follow fixed rules and cannot adapt when users phrase things differently. Because of this, many customer queries still end up unanswered or get routed to the wrong department.

This project aims to build a simple Customer Support Assistant that can automatically classify what a customer is asking about their intent and then pick the right response for it. We will use NLP to handle the text classification part, and a small Reinforcement Learning (RL) agent to learn which response works best over time.

The system focuses on English e-commerce customer support messages. When a customer types something like "where is my order" or "I want to return this item," the system predicts the intent, and selects a matching template response. This is an offline simulation and academic prototype, not a real deployed product.

1.2 SCOPE

- Text-based intent classification (single-turn customer messages)
- A contextual bandit RL agent that selects responses based on predicted intent
- A Text-CNN experiment as the CNN component
- Evaluation on a public e-commerce customer support dataset

1.3 SUCCESS METRICS

Table 1. Proposed Success Metrics

Metric	Target
Intent Macro-F1 (NLP classifier)	≥ 0.85 on test set
Task Success Rate (RL agent)	≥ 70% correct response selected
RL Cumulative Reward	At least 20% better than random policy
Text-CNN Macro-F1	≥ 0.80
Full pipeline runtime	≤ 90 minutes on Google Colab

2. RELATED WORK

Our system draws from prior work across intent classification, text neural networks, pre-trained language models, and reinforcement learning for dialogue.

Larson et al. [1] released CLINC150, a benchmark covering 150 intents across domains including e-commerce shopping. Their introduction of out-of-scope (OOS) detection, flagging queries that do not match any known intent directly motivated our confidence-threshold design, where low-confidence predictions are escalated rather than misrouted.

For the CNN component, we follow Kim [2], who showed that a shallow CNN over word embeddings using parallel filters of varying sizes achieves strong text classification results. Despite its simplicity, the Text-CNN remains a competitive and easy-to-implement baseline that we replicate for our CNN experiment.

For our core classifier, we draw from Devlin et al. [3], who introduced BERT, a bidirectional transformer pre-trained on large text corpora and fine-tuned on downstream tasks. We use DistilBERT, a lighter version of BERT, to keep training within our runtime constraint.

The RL component is based on Li et al. [4], who proposed LinUCB, a contextual bandit that balances exploration and exploitation via an upper confidence bound. Originally applied to news recommendation, LinUCB generalizes naturally to response selection: the agent observes intent embeddings and learns to pick the best response over many interactions.

Cui et al. [5] built SuperAgent, an e-commerce chatbot combining FAQ matching and retrieval. Their work highlights that customer queries are highly varied, reinforcing our choice of a fixed interpretable action space rather than free-form generation.

Finally, Williams et al. [6] showed through Hybrid Code Networks that pairing a learned NLP model with fixed domain-specific action templates outperforms fully generative end-to-end dialogue systems for goal-oriented tasks which directly justify our template-based response design.

3. DATASET PLAN

Primary: Bitext Customer Support LLM Chatbot Training Dataset

- **Source:** https://huggingface.co/datasets/bitext/Bitext-customer-support-llm-chatbot-training-dataset
- **License:** CC BY 4.0
- **Details:** ~26,872 utterances across 27 e-commerce intents (e.g., "track order," "request refund," "cancel order"). Synthetically generated with no PII. We apply a stratified 80/10/10 train/val/test split.
- **Limitation:** English-only and synthetic, which may not fully reflect real user language variation.

RL Environment: Simulated offline using the test split. Each sample is one episode the bandit selects a response and receives +1 if correct, 0 otherwise. Compared against random and greedy baselines.

4. PLANNED COMPONENTS

- **NLP - Intent Classifier (Core Model):** DistilBERT-base-uncased fine-tuned on the Bitext dataset. Input: tokenized message (max 128 tokens). Output: predicted intent from 27 classes. Training: AdamW, lr = 2e-5, 5 epochs, early stopping (patience = 2). Evaluation: macro-F1 and confusion matrix.
- **CNN – Text-CNN (Supporting Experiment):** Kim [2] architecture with filters of sizes [2,3,4] over GloVe-300d embeddings. Ablations: (A1) static vs. fine-tuned embeddings; (A2) filter sizes [2,3,4] vs. [3,4,5].
- **RL – Contextual Bandit:** LinUCB [4] with context = DistilBERT [CLS] embedding + confidence score. Action space = 27 response templates. Run over ≥ 3 seeds; report cumulative reward and regret curves.
- **MVP:** DistilBERT ≥ 0.85 macro-F1 + Text-CNN with both ablations logged + LinUCB beating random by ≥ 20%, all via a single bash run.sh within 90 minutes on Colab.

4. ETHICS RISK REGISTER

- **Risk 1 – Misclassification leading to wrong response (High / High)** A wrong prediction on a sensitive query like a refund dispute could frustrate users or cause financial confusion. Mitigation: confidence threshold of 0.70 — queries below it are flagged for human escalation. Confused intent pairs will be documented in the model card.
- **Risk 2 – Unequal performance across user groups (Medium / High)** Users who write informally or in short messages may get worse

predictions. Mitigation: slice analysis by utterance length and vocabulary diversity; data augmentation via paraphrasing for underperforming slices.

- **Risk 3 – RL agent neglecting rare intents (Medium / Medium)** Frequent intents dominate simulation rewards, causing the bandit to underserve rare but critical ones. Mitigation: separate reward tracking for high- vs. low-frequency intents; rule-based override for high-priority categories.

7. REFERENCES

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[2] Kim, Y. 2014. Convolutional Neural Networks for Sentence Classification. In *Proceedings of EMNLP 2014*. ACL, Doha, Qatar, 1746–1751. https://doi.org/10.3115/v1/D14-1181

[3] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL-HLT 2019*, Vol. 1. ACL, Minneapolis, Minnesota, 4171–4186. https://doi.org/10.18653/v1/N19-1423

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[5] Cui, L., Huang, S., Wei, F., Tan, C., Duan, C., and Zhou, M. 2017. SuperAgent: A Customer Service Chatbot for E-commerce Websites. In *Proceedings of ACL 2017 System Demonstrations*. ACL, Vancouver, Canada, 97–102. https://aclanthology.org/P17-4017

[6] Williams, J. D., Asadi, K., and Zweig, G. 2017. Hybrid Code Networks: Practical and Efficient End-to-End Dialog Systems with Domain-Specific Action Templates. In *Proceedings of ACL 2017*, Vol. 1. ACL, Vancouver, Canada, 665–675. https://doi.org/10.18653/v1/P17-1062

8. TEAM ROLES & MILESTONES

- **Project Lead / Integration:** Dizon Ghayzelle
- **Data & Ethics Lead:** Manangu, Redenelle
- **Modeling Lead:** Alvarado, Rafael
- **Evaluation & MLOps Lead:** Malig, Louis

Week 1 — Proposal & Setup

Proposal (2–3 pages in docs/):

- Problem statement & scope; success metrics and constraints
- Related work (3–6 citations)
- Dataset plan (source, license, consent, representativeness)
- Planned CNN/NLP/RL components and MVP
- Ethics risk register (top 3 risks + mitigations)
- Team roles & milestones

Other deliverables:

- GitHub repo initialized: README, environment, data download script/template, project board
- GitHub Release v0.1, Proposal PDF, and short 2–3 min screencast *(optional but recommended)*

Week 2 — Checkpoint

- Data acquired/cleaned; splits finalized; EDA notebook
- Baselines trained (simple ML + DL); initial metrics logged
- CNN experiment running (first results); NLP component scaffolded/prototyped
- RL agent stubbed with reward design; early learning curves (even if noisy)
- Draft Model Card sections started; Ethics statement updated
- Deliver: 1–2 page Checkpoint Report in docs/ + updated repo

Week 3 — Final Submission & Defense:

- Code frozen and reproducible (make repro or run.sh)
- Results tables/plots; ablations (≥ 2); error/slice analysis
- Final Report (4–6 pages) conference-style PDF
- Slide Deck (8–10 slides) and Live Demo *(or 3–5 min recorded demo)*